LEVEL 2 · 4 HOURS · APEX PRECISION MANUFACTURING

Measure a manufacturing Genie agent before you trust it.

A standalone lab for data engineers and solution architects. You compare table joins to certified measures, then score accuracy and latency.

10M telemetry | Metric views | Scorecard

4h	10M	25	12
Lab duration	Default telemetry rows	Questions per agent	IQ Scan checks

LAB JOURNEY

This lab turns a curated agent into a measured system.

1	2	3	4	5	6	7
Complex data	Metric views	A vs B agents	Scorecard	Tune layout	IQ Scan	Ask + Test Lab

Order. `10_complex_data` → `11_metric_views` → `12_agent_v2` → `13_accuracy_and_latency` → `14_performance_tuning` → Workbench → app.

SCALE THE MODEL

The Apex model adds production-scale grains and relationships.

Telemetry

10 million rows by default. Optional 50 million if the warehouse can take it.

SCD2 equipment

Line identity changes over time. Joins must pick the right version.

BOM and suppliers

Part explosion and vendor grain. Easy to double-count.

Work orders

Maintenance and production share keys. Grain is the trap.

CERTIFIED MEASURES

Metric views make the business grain explicit.

OEE

Availability × performance × quality. Always a ratio of sums, never an average of percents.

FPY

First-pass yield at the certified grain. Do not average line percents.

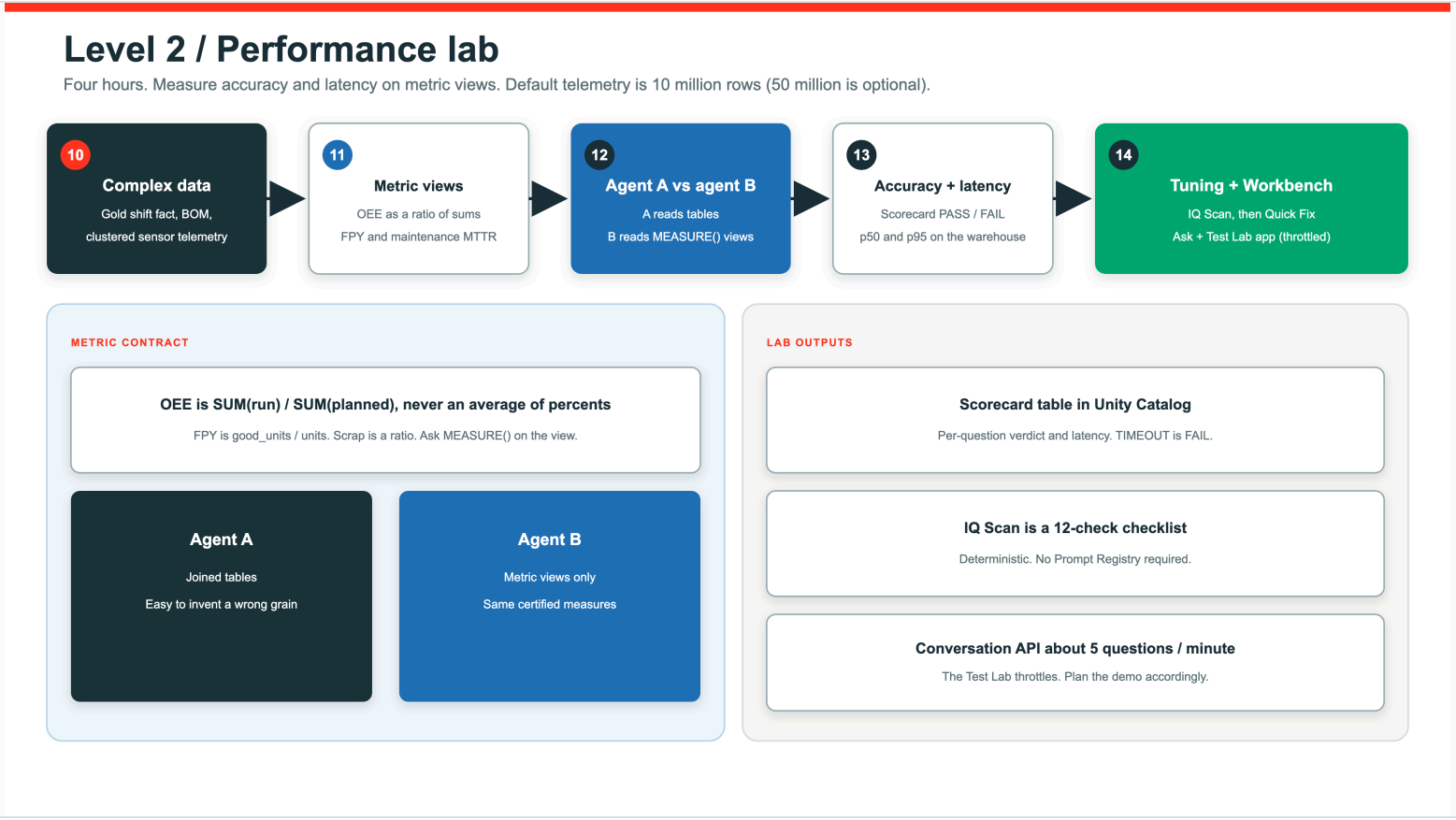
Maintenance

MTBF and related measures from work orders, not ad-hoc SQL in the agent.

Constraint. Materialized metric views and entity matching cannot combine with row filters, masks, or ABAC.

A VERSUS B

Agent A joins tables. Agent B calls certified measures.



SCORECARD

Twenty-five questions expose accuracy and latency together.

25	PASS/FAIL	p50	p95
Questions per agent	Exact match only	Median latency	Tail latency

TIMEOUT is FAIL. Do not tune the warehouse until both agents have a scorecard.

TUNE AFTER EVIDENCE

Tune data layout and compute only after the scorecard exists.

Liquid clustering

Cluster on the join and filter keys the scorecard actually uses.

Warehouse size

Move size only if p95 is the problem, not accuracy.

Query shape

Fix measures and grain before you add cache folklore.

Rerun evidence

Same 25 questions. Keep the before and after scorecard.

WORKBENCH

IQ Scan finds gaps. Quick Fix proposes a patch. A human applies it.

IQ Scan

12 checks on instructions, examples, joins, and benchmarks. Deterministic. No Prompt Registry.

Quick Fix

Suggested patches for failing checks. You read them. You apply them.

Human apply

Rescan, then rerun the 25-question scorecard. Nothing writes itself into prod.

Auto-Optimize is not part of Level 2. Skip any Prompt Registry path in this room.

CLOSE THE LAB

Take the measured agent into the Ask + Test Lab.

Reference app

Deploy `level-2-performance-lab/app/` or vibe-code from `vibe/PROMPT.md`.

Throttle

Conversation API is about 5 questions per minute. The app must throttle or you will see false TIMEOUTs.

This lab ends when the scorecard and the Test Lab agree. You do not need another lab to finish the day.